AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

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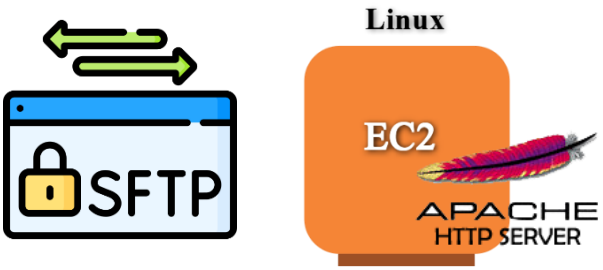
AWS Academy Certified Educator

7.Configure SFTP to Linux Web Server in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

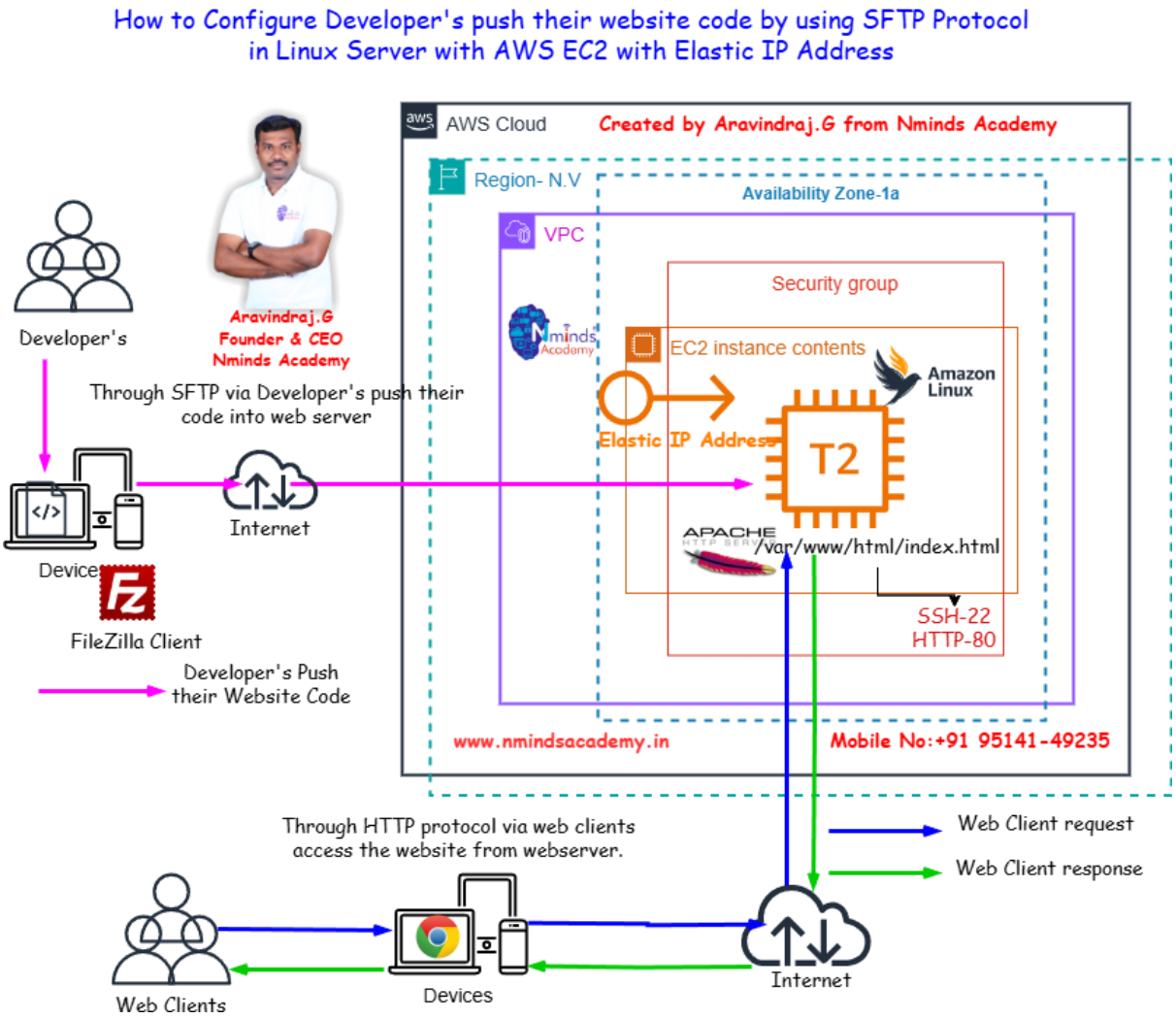
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.

5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology



Used Services in AWS

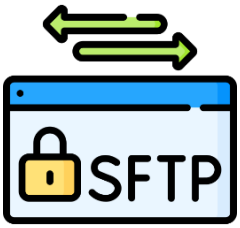
Used Services in Windows Server

Amazon EC2

APACHE HTTP SERVER

Apache Webserver

Amazon Elastic IP Address



Execution Tasks:

```
ec2-user@ip-172-31-25-88:~
~
~
~
~
~
lines 1-22/22 (END)
[1]+  Stopped                 sudo systemctl status sshd
[ec2-user@ip-172-31-25-88 ~]$ cat /etc/passwd | grep sftpuser
sftpuser:x:1001:1001::/home/sftpuser:/bin/bash
[ec2-user@ip-172-31-25-88 ~]$ sudo passwd sftpuser
Changing password for user sftpuser.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-25-88 ~]$ ls -ld /home/sftpuser
drwx------. 2 sftpuser sftpuser 76 Mar  9 15:44 /home/sftpuser
[ec2-user@ip-172-31-25-88 ~]$ sudo nano /etc/ssh/sshd_config
[ec2-user@ip-172-31-25-88 ~]$ sudo systemctl restart sshd
[ec2-user@ip-172-31-25-88 ~]$ sftp sftpuser@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:ANioxJ+jdIH7OKSitFlSa3/HPnqqhOA7J+1oIkDSCfM.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
sftpuser@localhost's password:
Connected to localhost.
sftp> exit
[ec2-user@ip-172-31-25-88 ~]$ ls /home/sftpuser
ls: cannot open directory '/home/sftpuser': Permission denied
[ec2-user@ip-172-31-25-88 ~]$ sudo ls /home/sftpuser
index.html
[ec2-user@ip-172-31-25-88 ~]$
```